

AI-powered customer profile deduplication

Entity resolution prototype for a discount marketplace

A supervised matching and graph-clustering pipeline that turns noisy customer records into actionable merge decisions.

0.962

precision at auto-merge threshold

96.75%

true entities not split after clustering

Executive summary

A practical entity resolution prototype for cleaner customer data and safer automation

Business issue

Duplicate profiles fragment purchase history, loyalty rewards, personalization signals, and core analytics such as LTV, retention, RFM, and cohorts.

AI solution

Build a pairwise matching model with engineered similarity features, then convert high-confidence matches into customer clusters using connected components.

Operating model

Use a high threshold for automatic merges, a manual-review band for uncertain cases, and audit / rollback controls to reduce business risk.

Case outcome

0.87

F1 at conservative auto-merge threshold

8,224

predicted clusters on validation

1,093

high-confidence auto-merge edges

0.997

NMI cluster quality

Why this matters commercially

Duplicate customers create compounding losses across personalization, loyalty, fraud, and analytics

Personalization

Fragmented behavior weakens recommendations and campaign targeting.

Loyalty cost

One customer can receive duplicated bonuses, discounts, or loyalty cards.

Analytics trust

LTV, retention, RFM, and cohort analysis are distorted at the customer level.

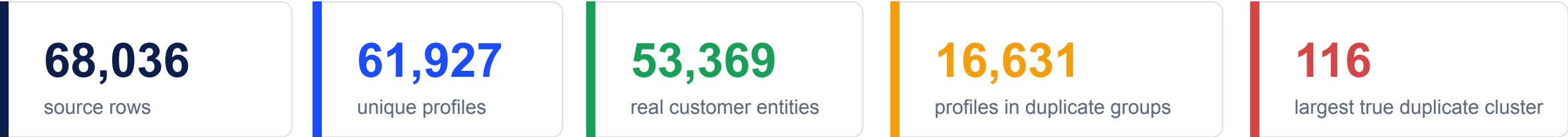
Fraud surface

Multiple accounts create additional vectors for abuse and policy gaming.

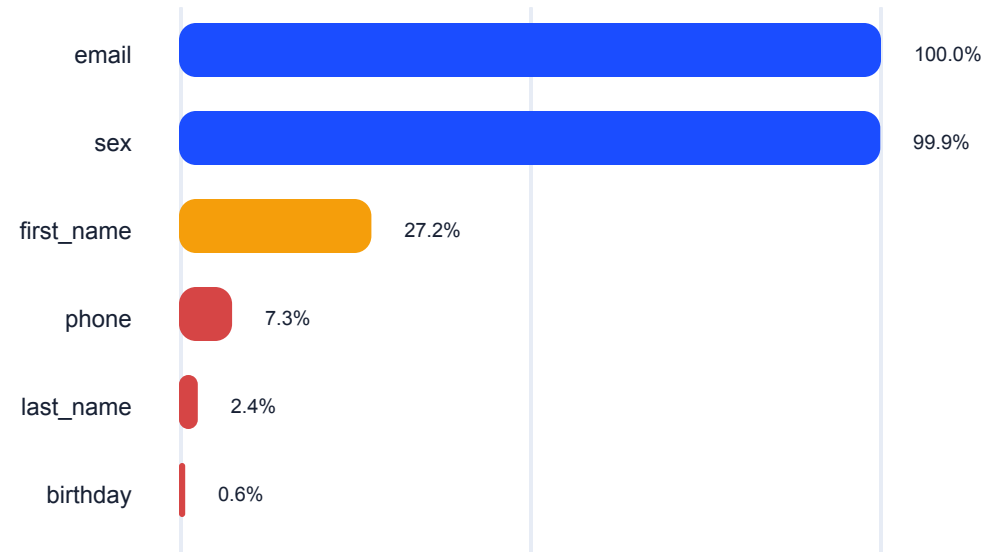
Consulting implication: solve identity resolution as a controlled decisioning system, not as a one-off exact-match rule.

Data reality: meaningful duplicates, sparse identity data

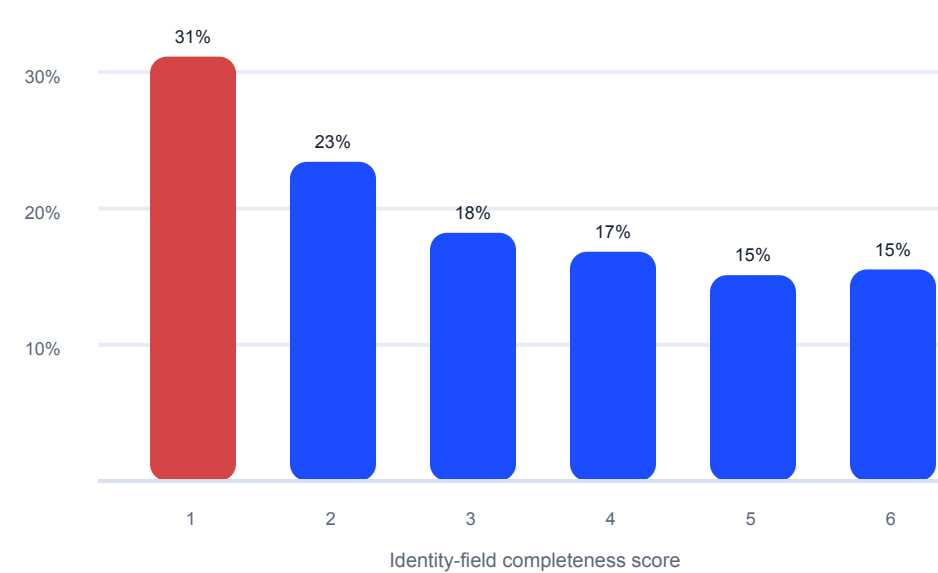
Exact matching is insufficient because critical identity fields are missing or noisy



Field completeness after profiling



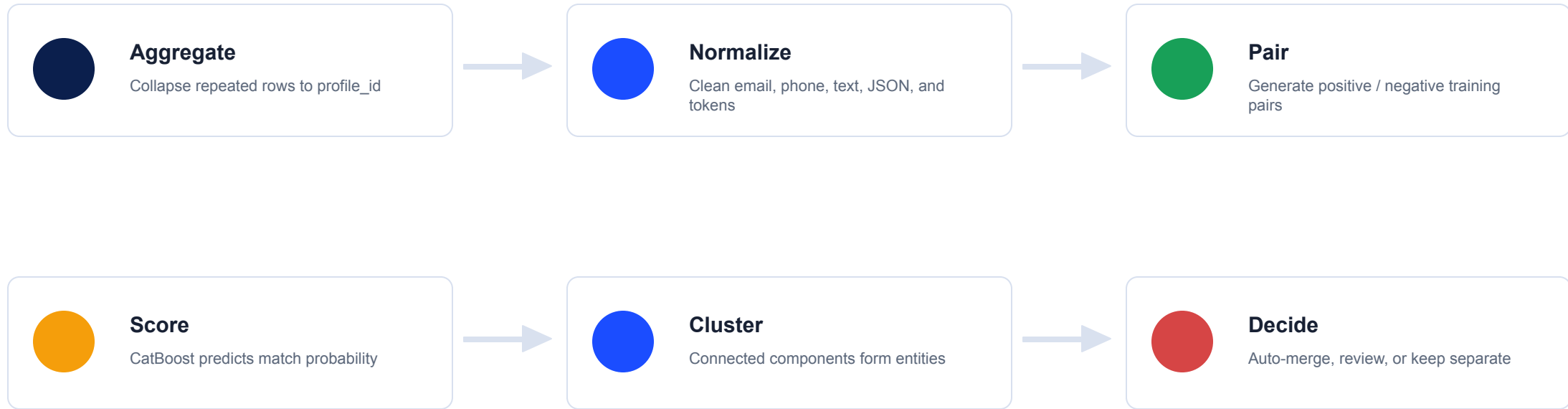
Duplicate rate falls as profile completeness rises



Key insight: duplicates concentrate among sparse profiles, so the model must combine identity, technical, behavioral, geographic, and temporal signals.

Solution architecture

From raw profiles to governed merge decisions



Design principle: maximize precision for automated actions and route uncertain cases to human review.

Feature strategy

The model uses multiple weak and strong signals rather than relying on one brittle identifier

Contact

phone match, email-domain match, email local-part similarity

Identity

first name, full-name similarity, birth year, known-sex conflict

Geography

city, region, country, geoid, timezone

Technical

device, browser, OS, non_processing_features Jaccard

Behavioral

fs_features Jaccard and behavior_mean_sim

Temporal & completeness

creation-time gap, exact-match count, sparse-profile flags

Signal hierarchy

High precision: same normalized phone

Broad coverage: email local-part similarity

Sparse-profile lift: behavioral + technical tokens

A Decision Tree baseline was fragile because it relied mainly on behavior_mean_sim; CatBoost distributed importance across signals.

Model performance and decision thresholds

Thresholding translates probabilities into business-safe actions

Pairwise validation



Decision policy

Score	Decision	Business action
>= 0.85	auto_merge	High-confidence merge; transfer history, loyalty, and preferences
0.60-0.85	manual_review	Uncertain pair or cluster; support team confirms
< 0.60	do_not_merge	Keep profiles separate

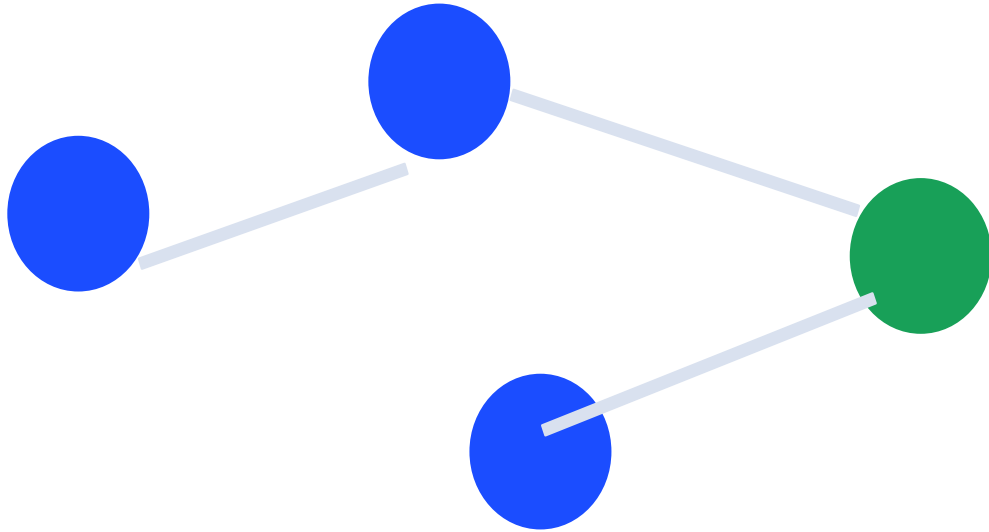
At 0.85 threshold

Precision
0.962
Recall 0.794
F1 0.870

Clustering outcome

Graph components convert pairwise matches into customer entities

How it works



Profiles are graph vertices;
high-confidence matches are edges.

Validation results

9,266

validation profiles / graph
vertices

1.127

mean predicted cluster
size

0.864

ARI

0.997

NMI

**96.75% of true entities were reconstructed
without being split across clusters.**

Prototype evidence

Demonstrated in this case

Anonymized synthetic data | offline validation

- 1 Reproducible entity-resolution pipeline built end to end
- 2 Pairwise matching and graph clustering evaluated on held-out data
- 3 Conservative thresholds proposed for merge, review, and no-merge decisions
- 4 Error analysis prioritizes avoiding harmful false merges

Not yet demonstrated

Requires real-world testing before deployment

- ? Generalization to live customer data and changing duplicate patterns
- ? Realized revenue uplift, loyalty savings, or fraud reduction
- ? Production scalability, latency, monitoring, and rollback performance
- ? Safety and support workload of automated merge decisions

Recommended next step

Run a shadow-mode pilot on fresh labeled data; quantify false-merge cost and review workload before enabling automation.